

HOMEWORK 5

Due date: Monday of Week 6

Exercises: 1, 2, 3, 6, 11, 14, pages 288-290

Exercises: 1, 2, 4, 5, 6, 7, 10, 12, pages 298-299.

Problem 1. Let m be any fixed positive integer. Show that

$$\sum_{n=1}^m \frac{1}{n^2} \leq \frac{\pi^2}{6},$$

and

$$\sum_{n=1}^m \frac{1}{n^4} \leq \frac{\pi^4}{90}.$$

Try to generalize the above to $\sum_{n=1}^m \frac{1}{n^{2k}}$ for $k = 3, 4$.

Let $F = \mathbb{R}$ or $\mathbb{C}$. Let $(V, +, \cdot)$ be a vector space over F , where $+$: $V \times V \rightarrow V$ is the addition and $\cdot$: $F \times V \rightarrow V$ is the scalar product. Define a new vector space $(\bar{V}, +, \odot)$ by

- $(\bar{V}, +) = (V, +)$ as an abelian group;
- the new scalar product $\odot$: $F \times V \rightarrow V$ is defined to be $c \odot \alpha := \bar{c} \cdot \alpha$, for all $c \in F, \alpha \in V$.

Note that, if $F = \mathbb{R}$, then $\bar{V}$ is the same as V .

Problem 2. (1) Show that $(\bar{V}, +, \odot)$ is indeed a vector space.

(2) If $\dim V$ is finite, then $\dim \bar{V} = \dim V$.

(3) Let $T : V \rightarrow V$ be a linear map, show that $T : \bar{V} \rightarrow \bar{V}$ is also a linear map.

Problem 3. In Theorem 7, page 293 of the textbook, we only defined adjoint for linear operators $T \in \text{End}(V)$. Try to generalize this concept to general linear maps, namely, try to define the adjoint T^* for $T \in \text{Hom}(V, W)$, where V, W are two (possibly different) inner product spaces over F . Moreover, show that the adjoint you defined above indeed exists.

Problem 4. Let $A \in \text{Mat}_{m \times n}(F)$. We consider the linear operator $T_A : F^n \rightarrow F^m$ by $T_A(\alpha) := A\alpha$. Here F^n and F^m are viewed as inner product space with respect to the standard inner product defined on them. Show that the adjoint of T_A is given by T_{A^*} , where $A^* = \bar{A}^t$ and the adjoint of T_A is defined in the last problem.

We consider the column vector space $\mathbb{R}^n$ over $\mathbb{R}$. Let $(\cdot | \cdot)$ be the standard inner product on $\mathbb{R}^n$. Recall that $(x|y) = y^t x$, where y^t denotes the transpose of y . Given a matrix $A \in \text{Mat}_{m \times n}(\mathbb{R})$, we define the linear operator $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ by $T_A(x) = Ax$. Denote $\text{Ker}(A) = \{x \in \mathbb{R}^n : Ax = 0\} = \text{Ker}(T_A)$. Let $\text{Row}(A)$ denote the space spanned by rows of A .

Problem 5. Given $A \in \text{Mat}_{m \times n}(\mathbb{R})$. Show that

- (1) $\text{Ker}(A) = \text{Ker}(A^t A)$;
- (2) $\text{rank}(A) = \text{rank}(A^t A)$;
- (3) $\text{Row}(A) = \text{Row}(A^t A)$.

Let F be $\mathbb{R}$ or $\mathbb{C}$. Let V be a finite dimensional inner product space over F and let W be a subspace of V . We then have the orthogonal decomposition $V = W \oplus W^\perp$. Let $\text{Proj}_W : V \rightarrow V$ denotes the projection from V to W corresponding to this decomposition, namely, $\text{Proj}_W(\alpha, \beta) = \alpha$ for $\alpha \in W, \beta \in W^\perp$.

Problem 6. Let $V = \mathbb{R}^n$ endowed with the standard inner product. Let $W \subset V$ be a subspace of V of dimension m . Let $\mathcal{B} = \{\alpha_1, \dots, \alpha_m\}$ be a basis of W and consider the matrix

$$M_{\mathcal{B}} = [\alpha_1, \dots, \alpha_m] \in \text{Mat}_{n \times m}(\mathbb{R}).$$

Here each α_i is a column vector.

- (1) Show that $M_{\mathcal{B}}^t M_{\mathcal{B}} \in \text{Mat}_{m \times m}(\mathbb{R})$ is invertible.
- (2) Consider the matrix $P_{\mathcal{B}} = M_{\mathcal{B}}(M_{\mathcal{B}}^t M_{\mathcal{B}})^{-1} M_{\mathcal{B}}^t \in \text{Mat}_{n \times n}(\mathbb{R})$. Show that $P_{\mathcal{B}}$ is independent on the choice of $\mathcal{B}$ and thus it only depends on the space W .
- (3) For any $\alpha \in \mathbb{R}^n$, show that $P_{\mathcal{B}}\alpha \in W$. (The notation $P_{\mathcal{B}}\alpha$ denotes the matrix product of $P_{\mathcal{B}}$ with α).
- (4) Show that the map $E : \mathbb{R}^n \rightarrow \mathbb{R}^n$ defined by $E(\alpha) = P_{\mathcal{B}}\alpha$ is the same as the projection map Proj_W . (Hint: one way to do this is by choosing a good basis of W using the last part. Then compute the matrix $P_{\mathcal{B}}$.)

Given a matrix $A \in \text{Mat}_{m \times n}(\mathbb{R})$ and $\beta \in \mathbb{R}^m$, we consider the linear system

$$(0.1) \quad Ax = \beta.$$

The above equation does not always have a solution. If the above equation has no solution, we can consider the following approximating solution, which is called *least square solution*. A vector $\hat{x} \in \mathbb{R}^n$ is called a least square solution of (0.1) if

$$\|\beta - A\hat{x}\| \leq \|\beta - Ax\|, \forall x \in \mathbb{R}^n.$$

It is clear that if $x \in \mathbb{R}^n$ is a solution of (0.1), then it is also a least square solution.

Problem 7. Given $A \in \text{Mat}_{m \times n}(\mathbb{R})$ and $\beta \in \mathbb{R}^m$. Consider

$$\text{Im}(A) = \{y \in \mathbb{R}^m : y = Ax \text{ for some } x \in \mathbb{R}^n\} \subset \mathbb{R}^m,$$

and

$$\text{Ker}(A^t) = \{y \in \mathbb{R}^m : A^t y = 0\} \subset \mathbb{R}^m.$$

- (1) Show that $\text{Im}(A)^\perp = \text{Ker}(A^t)$. Here $\perp$ is relative to the standard inner product on $\mathbb{R}^m$.
- (2) Show that $\hat{x} \in \mathbb{R}^n$ is a least square solution of (0.1) if and only if $\beta - A\hat{x} \in \text{Im}(A)^\perp$ if and only if $A^t A\hat{x} = A^t \beta$.
- (3) Show that (0.1) always has a least square solution.
- (4) Give a condition such that (0.1) has a unique least square solution.

Problem 8. Let $A = \begin{bmatrix} -1 & 3 & 2 \\ 2 & 1 & 3 \\ 0 & 1 & 1 \end{bmatrix} \in \text{Mat}_{3 \times 3}(\mathbb{R})$ and $\beta = \begin{bmatrix} 7 \\ 0 \\ 7 \end{bmatrix}$. Find a least square solution of the equation

$$Ax = \beta.$$

Problems 5-8 were stated for the field $\mathbb{R}$. Try to consider the analogues for inner product spaces over $\mathbb{C}$. For example, in Problem 3, if we replace $\mathbb{R}$ by $\mathbb{C}$ and replace A^t by A^* , then show the same assertions hold. In Problem 4, what is the matrix $P_{\mathcal{B}}$ if the field is $\mathbb{C}$? Formulate and solve the least square solution problem over $\mathbb{C}$ as in Problem 5.

Problem 9. Let $F = \mathbb{R}$ or $\mathbb{C}$. Let $V = \text{Mat}_{m \times n}(F)$ and consider the map

$$(\mid) : V \times V \rightarrow F$$

defined by $(A \mid B) = \text{tr}(AB^*)$, where $B^* = \overline{B}^t$. Show that $(\mid)$ is an inner product on V .

Given a matrix $A = (a_{i,j}) \in \text{Mat}_{n \times n}(\mathbb{C})$, consider the Hilbert-Schmidt norm which is defined by

$$\|A\|_{HS} = \sqrt{\text{tr}(AA^*)} = \sqrt{\sum_{i,j} |a_{i,j}|^2}.$$

Problem 10. *Show that*

$$\|A + B\|_{HS} \leq \|A\|_{HS} + \|B\|_{HS}$$

for all $A, B \in \text{Mat}_{n \times n}(\mathbb{C})$.